Lagrange's linear PDE : $pP + qQ = R$

Method of multipliers:

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{l\,dx + m\,dy + n\,dz}{lP + mQ + nR} = \frac{Nr}{0}$$

such that $lP + mQ + nR = 0$, then $= \frac{Nr}{0}$

$$\therefore\ l\,dx + m\,dy + n\,dz = 0$$

$$\Rightarrow\ lx + my + nz = c_1$$

Ex: Solve $x(y^2 - z^2)p - y(z^2 + x^2)q = z(x^2 + y^2)$

Ans: A.E. is $\dfrac{dx}{x(y^2 - z^2)} = \dfrac{dy}{-y(z^2 + x^2)} = \dfrac{dz}{z(x^2 + y^2)}$ — (1)

Taking x, y, z as multipliers, (1) becomes

$$\frac{x\,dx + y\,dy + z\,dz}{x^2(y^2 - z^2) - y^2(z^2 + x^2) + z^2(x^2 + y^2)} = \frac{x\,dx + y\,dy + z\,dz}{0}$$

$$\Rightarrow\ x\,dx + y\,dy + z\,dz = 0.$$

Integrating, $\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = c_0 \Rightarrow \boxed{x^2 + y^2 + z^2 = c_1}$

Again taking $\frac{1}{x}, -\frac{1}{y}, -\frac{1}{z}$ as multipliers, we get

$$\frac{\frac{1}{x}dx - \frac{1}{y}dy - \frac{1}{z}dz}{y^2 - z^2 + z^2 + x^2 - x^2 - y^2} = \frac{\frac{1}{x}dx - \frac{1}{y}dy - \frac{1}{z}dz}{0}$$

$\Rightarrow$ Integrating, $\ln x - \ln y - \ln z = \ln c_2$

$$\Rightarrow \ln\left(\frac{x}{yz}\right) = \ln c_2 \Rightarrow \frac{x}{yz} = c_2$$

$\therefore$ The required soln is $\phi\left(x^2+y^2+z^2, \frac{x}{yz}\right) = 0$

Solve: (1) $(y-2x)p + (x+yz)q = x^2+y^2$

Ans: $\phi(x^2-y^2, xy-z) = 0$

(2) $x(y^2+z)p - y(x^2+z)q = z(x^2-y^2)$

Ans: $\phi(x^2+y^2-2z, xyz) = 0$

(3) $(x+2z)p + (xz-y)q = 2x^2+y$

Ans: $\phi(xy-z^2, x^2-y-z) = 0$

(4) $(z^2-2yz)p + (xy+zx)q = xy-zx$

Ans: $\phi(y^2-z^2-2yz, x^2+y^2+z^2) = 0$

(5) Solve $(y^3x-2x^4)p + (2y^4-x^3y)q = 9z(x^3-y^3)$

Ans: A.E. $\dfrac{dx}{y^3x-2x^4} = \dfrac{dy}{2y^4-x^3y} = \dfrac{dz}{9z(x^3-y^3)}$ — (1)

Taking $\frac{1}{x}, \frac{1}{y}, \frac{1}{3z}$ as multipliers, (1) becomes

$$\frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{3z}dz}{y^3 - 2x^3 + 2y^3 - x^3 + 3x^3 - 3y^3} = \frac{\frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{3z}dz}{0}$$

$$\Rightarrow \frac{1}{x}dx + \frac{1}{y}dy + \frac{1}{3z}dz = 0$$

Integrating,

$$\ln x + \ln y + \frac{1}{3}\ln z = \ln c_1$$

$$\Rightarrow \ln(xyz^{1/3}) = \ln c_1 \Rightarrow \boxed{xyz^{1/3} = c_1}$$

Taking first two fractions of ①,

$$\frac{dx}{y^3x - 2x^4} = \frac{dy}{2y^4 - x^3y}$$

$$\Rightarrow (2y^4 - x^3y)dx = (xy^3 - 2x^4)dy \quad \text{—— } (*)$$

$$\Rightarrow \frac{dx}{dy} = \frac{xy^3 - 2x^4}{2y^4 - x^3y} \quad \text{—— } (2)$$

Let $y = vx \therefore \frac{dy}{dx} = v.1 + x\frac{dv}{dx}$

$$(2) \Rightarrow v + x\frac{dv}{dx} = \frac{x\,v^3x^3 - 2x^4}{2v^4x^4 - 2x^4} = \frac{v^3 - 2}{2v^4 - 2}$$

$$\Rightarrow x\frac{dv}{dx} = \frac{v^3 - 2}{2v^4 - 2} - v$$

$$= \frac{v^3 - 2 - 2v^5 + 2v}{2v^4 - 2}$$

$$\boxed{\frac{2v^4 - 2}{2v^5 + v^3 + 2v - 2}\,dv = \frac{1}{x}dx} \rightarrow \text{Integration of L.H.S is difficult.}$$

Here may apply exactness condition.

$$\therefore \frac{\partial M}{\partial y} = 8y^3 - x^3, \quad \frac{\partial N}{\partial x} = -y^3 + 8x^3$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

* is a homogeneous ODE

$$I.F = \frac{1}{Mx+Ny} = \frac{1}{y^3x^2 - 2x^5 - 2y^5 - x^3y^2}$$

<u>Alternative:</u>

$$\frac{dx}{y^3x - 2x^4} = \frac{dy}{2y^4 - x^3y}$$

$$\Rightarrow (2y^4 - x^3y)dx = (y^3x - 2x^4)dy$$

Dividing by x^3y^3,

$$\Rightarrow \left(\frac{2y}{x^3} - \frac{1}{y^2}\right)dx = \left(\frac{1}{x^2} - \frac{2x}{y^3}\right)dy$$

$$\Rightarrow \left(\frac{2y}{x^3} - \frac{1}{y^2}\right)dx - \left(\frac{1}{x^2} - \frac{2x}{y^3}\right)dy = 0$$

$$\Rightarrow -\left(\frac{1}{x^2}dy - \frac{2y}{x^3}dx\right) - \left(\frac{1}{y^2}dx - \frac{2x}{y^3}dy\right) = 0$$

$$\Rightarrow \left(\frac{1}{x^2}dy - \frac{2y}{x^3}dx\right) + \left(\frac{1}{y^2}dx - \frac{2x}{y^3}dy\right) = 0$$

$$\Rightarrow \int d\left(\frac{y}{x^2}\right) + \int d\left(\frac{x}{y^2}\right) = 0$$

$$\Rightarrow \frac{y}{x^2} + \frac{x}{y^2} = c_2$$

The required soln $\phi\left(xyz^{1/3}, \frac{y}{x^2} + \frac{x}{y^2}\right) = 0$

* Solve $x(x^2+3y^2)p - y(3x^2+y^2)q = 2z(y^2-x^2)$

<u>Ans:</u> A.E is $\frac{dx}{x(x^2+3y^2)} = \frac{dy}{-y(3x^2+y^2)} = \frac{dz}{2z(y^2-x^2)}$

Taking $\frac{1}{x}, \frac{1}{y}, -\frac{1}{z}$ as multipliers, ① becomes

$$\frac{\frac{1}{x}dx+\frac{1}{y}dy-\frac{1}{z}dz}{x^2+3y^2-3x^2-y^2-2y^2+2x^2} = \frac{\frac{1}{x}dx+\frac{1}{y}dy-\frac{1}{z}dz}{0}$$

$$\Rightarrow \frac{1}{x}dx+\frac{1}{y}dy-\frac{1}{z}dz=0$$

$\Rightarrow$ Integrating, $\ln x+\ln y-\ln z=\ln c_1$

$$\Rightarrow \ln\left(\frac{xy}{z}\right)=\ln c_1 \Rightarrow \boxed{\frac{xy}{z}=c_1}$$

Taking 1st two fractions,

$$\frac{dx}{x(x^2+3y^2)}=\frac{dy}{-y(3x^2+y^2)}$$

$$\Rightarrow \frac{dy}{dx}=-\frac{y(3x^2+y^2)}{x(x^2+3y^2)} \quad \text{——— (*)}$$

Let $y=vx \Rightarrow \frac{dy}{dx}=v+x\frac{dv}{dx}$

$$(*) \Rightarrow v+x\frac{dv}{dx}=\frac{-vx(3x+v^2x^2)}{x(x^2+3\cdot v^2x^2)}=\frac{-3v-v^3}{1+3v^2}$$

$$\Rightarrow x\frac{dv}{dx}=\frac{-3v-v^3}{1+3v^2}-v=\frac{-3v-v^3-v-3v^3}{1+3v^2}$$

$$\Rightarrow x\frac{dv}{dx}=\frac{-4v-4v^3}{1+3v^2}=\frac{4v(1+v^2)}{1+3v^2}$$

$$\Rightarrow \int\frac{3v^2+1}{v(1+v^2)}=\int\frac{1}{4x}dx \Rightarrow \int\left(\frac{1}{v}+\frac{2v}{1+v^2}\right)dv=\frac{1}{4}\ln x+\ln(c_2)$$

$$\Rightarrow \ln v+\ln(1+v^2)=\ln x^{1/4}+\ln c_2$$

$$\Rightarrow \ln\{v(1+v^2)\}=\ln x^{1/4}+\ln c_2$$

$$\Rightarrow \ln\left[\frac{y}{x}\left(1+\frac{y^2}{x^2}\right)\right] - \ln x^{1/4} = \ln c_2$$

$$\Rightarrow \ln \frac{\frac{y}{x}\left(1+\frac{y^2}{x^2}\right)}{x^{1/4}} = \ln c_2$$

$$\Rightarrow \frac{\frac{y}{x}\left(1+\frac{y^2}{x^2}\right)}{x^{1/4}} = c_2$$

$\therefore$ The required soln is $\left(\frac{xy}{z}, \frac{\frac{y}{x}\left(1+\frac{y^2}{x^2}\right)}{x^{1/4}}\right) = 0$

Solve (5) $(y-z)p+(z-x)q = x-y$

Ans: $\phi(x+y+z, x^2+y^2+z^2) = 0$

(6) $(y+zx)p-(x+yz)q+y^2-x^2=0$

Ans. $\phi(x^2+y^2-z^2, xy+z) = 0$

(7) $x(y-z)p+y(z-x)q = z(x-y)$

Ans: $\phi(x+y+z, xyz) = 0$.

(8) $x^2p+y^2q = (x+y)z$

Ans: $\phi\left(\frac{1}{y}-\frac{1}{x}, \frac{xy}{z}\right) = 0$.